

ORR, MN, USA

WMO: 726544

Lat: 48.016N

Lon: 92.856W

Elev: 400

StdP: 96.62

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04958

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -31.8	(c) -28.0	(d) -35.5	(e) 0.1	(f) -31.4	(g) -32.2	(h) 0.2	(i) -27.8	(j) 8.7	(k) -12.3	(l) 8.0	(m) -12.4	(n) 1.4	(o) 300	(p) 0.580

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	29.2	20.8	27.7	19.7	26.4	18.7	23.0	27.1	21.7	25.9	20.4	24.5	3.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	17.6	26.2	20.2	15.7	24.1	18.9	14.3	22.9	70.6	26.7	65.0	26.2	60.7	25.1	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.2	7.2	6.0	-34.9	32.4	3.5	1.9	-37.5	33.7	-39.5	34.8	-41.4	35.9	-44.0	37.3
			-34.8	24.6	3.4	1.4	-37.3	25.6	-39.3	26.5	-41.3	27.3	-43.8	28.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.7	-14.0	-11.4	-4.0	3.7	10.9	16.1	18.8	17.6	13.4	6.1	-2.8	-10.5		
	DBStd	12.89	8.26	7.81	7.46	5.65	4.88	3.91	3.27	3.39	4.85	5.06	6.37	7.45		
	HDD10.0	3214	744	600	437	201	47	3	0	1	21	143	386	634		
	HDD18.3	5445	1002	833	693	440	236	87	33	55	159	379	635	892		
	CDD10.0	932	0	0	2	10	74	187	274	237	123	23	1	0		
	CDD18.3	119	0	0	0	0	5	21	49	32	12	1	0	0		
	CDH23.3	1293	0	0	1	6	96	236	508	317	122	9	0	0		
	CDH26.7	300	0	0	0	1	22	52	128	69	25	2	0	0		
(14) Wind	WSAvg	2.2	2.3	2.2	2.3	2.3	2.4	2.0	1.9	1.8	2.1	2.5	2.5	2.1		
(15) Precipitation	PrecAvg	706	22	17	28	50	76	101	102	81	94	65	38	28		
	PrecMax	854	50	34	85	111	130	282	204	157	188	110	87	59		
	PrecMin	588	9	3	10	15	29	44	42	18	30	15	2	14		
	PrecStd	78	11	9	17	22	27	51	47	37	46	28	22	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	7.7	17.4	22.9	29.0	31.2	32.5	31.2	29.1	24.0	14.1	5.3		
		MCWB	1.6	3.1	10.1	12.9	16.7	22.3	23.1	21.9	20.3	15.4	9.6	2.4		
	2%	DB	1.4	4.1	12.4	19.1	26.2	27.7	29.3	28.1	27.1	20.1	11.3	2.7		
		MCWB	0.1	1.5	7.1	10.3	15.5	19.4	21.7	20.2	19.4	13.8	8.3	1.1		
	5%	DB	-0.9	2.2	8.7	17.1	22.9	26.2	27.7	27.1	24.0	17.2	7.9	1.2		
		MCWB	-1.5	0.5	4.7	8.6	13.5	18.2	20.3	19.5	18.1	12.2	5.4	0.1		
	10%	DB	-2.5	-0.1	6.1	13.7	21.0	23.9	26.3	24.9	22.2	13.9	6.1	-0.9		
		MCWB	-3.3	-1.3	2.6	7.1	13.1	17.3	19.2	18.6	16.9	10.2	3.7	-1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.4	4.4	12.7	14.3	20.2	24.0	25.0	24.0	23.4	18.1	12.2	3.0		
		MCDB	4.2	6.9	14.7	20.5	24.9	28.2	29.7	28.2	28.0	20.1	13.2	4.4		
	2%	WB	0.3	2.3	7.8	11.6	17.5	21.7	23.3	21.9	21.1	15.0	8.0	1.3		
		MCDB	1.3	4.0	11.1	17.0	22.6	26.3	27.6	26.2	24.6	18.6	10.2	2.6		
	5%	WB	-1.4	0.5	5.1	9.5	15.5	20.1	22.0	20.7	19.1	12.7	5.9	0.0		
		MCDB	-0.7	2.0	8.5	15.5	21.0	23.8	26.3	25.3	22.5	16.7	8.0	0.9		
	10%	WB	-3.2	-1.6	2.9	7.6	13.8	18.6	20.5	19.5	17.5	10.9	3.7	-1.3		
		MCDB	-2.5	-0.3	5.6	12.6	18.7	22.3	24.6	23.4	20.7	13.4	5.8	-0.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.7	11.9	12.4	13.2	12.4	12.7	12.5	11.9	9.4	7.9	8.5		
		MCDBR	9.5	11.6	13.7	17.6	17.6	15.0	14.4	14.5	14.8	14.0	10.9	9.0		
	5% WB	MCWBR	8.2	8.8	8.9	10.2	8.9	7.9	7.5	7.4	8.3	8.4	7.6	7.4		
		MCWBR	9.0	9.9	12.4	15.4	14.4	12.5	13.2	12.5	12.4	12.5	10.3	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.252	0.266	0.315	0.341	0.357	0.372	0.387	0.372	0.356	0.306	0.282	0.260		
		taud	2.360	2.327	2.343	2.395	2.407	2.384	2.361	2.436	2.452	2.581	2.519	2.394		
	Ebn at Noon		840	904	904	911	906	890	870	870	850	855	804	783		
		Edh at Noon	68	88	104	109	112	116	117	104	93	70	59	58		
(44) All-Sky Solar Radiation	RadAvg		1.36	2.52	3.73	4.71	5.25	5.58	5.95	5.04	3.73	2.19	1.23	0.96		
	RadStd		0.17	0.25	0.35	0.45	0.36	0.43	0.30	0.36	0.30	0.30	0.16	0.10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	-1.00	-1.52	N/A	N/A	N/A
(47) Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page